TUTORIAL MODULE | SC025 | 2026/2027



CHAPTER 1
INTRODUCTION TO PROGRAMMING

NAME:
CLASS:

1. Look at these two pieces of code [5m]:

A:	**B:**
CLC LDX #0 loop: LDA A, X ADC B, X STA C, X NEXT INX CPX #16 BNE loop	FOR loop = 1 TO 4 INPUT Number1, Number2 Sum = Number1 + Number2 PRINT Sum

a. Which of these pieces of code is written in a high-level language?

_____

b. Give one benefit of writing code in a high-level language.

_____

c. Give one benefit of writing code in a low-level language.

_____
_____
_____

d. High-level languages can be compiled or interpreted.
   i. Give two differences between a compiler and an interpreter.

Compiler	**Interpreter**

2. Explain the following terms in the context of programming. (3m)
    a. Program:

    _____
    _____
    _____

    b. Programming language:

    _____
    _____
    _____

    c. Programming paradigm:

    _____
    _____
    _____

3. Explains the following terms as used in the programming paradigm. [3m]
    a. Object Oriented:

    _____
    _____

    b. Logic:

    _____
    _____

    c. Procedural:

    _____
    _____

4. Programming is a process that leads from an original formulation of a computing problem to executable computer programs.
    a. Give **one (1)** difference between compiler and interpreter. [2 *marks*]

Compiler	Interpreter

b. Explain **one (1)** disadvantages of Assembly Language. [2 *marks*]

_____
_____
_____

c. State **two (2)** examples of High Level Languages other than C++. [2 *marks*]

_____
_____

5.

> *A programming language is a computer language engineered to create a standard form of commands. These commands can be interpreted into a code understood by a machine. Programs are created through programming languages to control the behavior and output of a machine through accurate algorithms, similar to the human communication process.*
> https://www.techopedia.com/definition/24815/programming-language

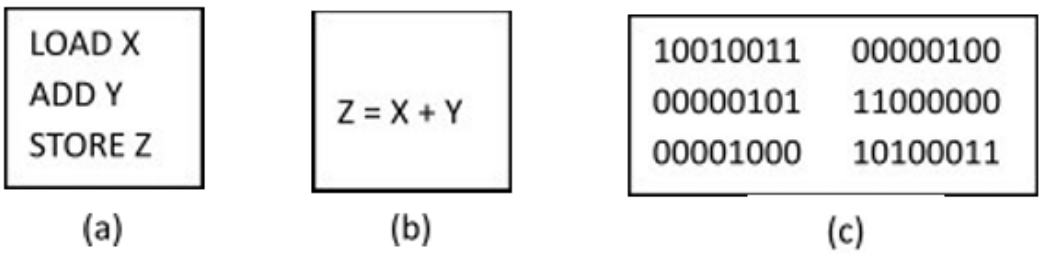
a. Give one (1) type of programming language (1m)

_____
_____

b. What is meant by the underlined sentence in the above statement (2m)

_____
_____

6. Figure shows three program segments written in three different languages. The program is to calculate the sum of X and Y and put the value into Z.



a. State the translator needed to convert the program :
   i. From (a) to (c)

_____
_____

ii. From (b) to (c)

_____

_____

7. Programming paradigm is a general framework or model for designing and organizing a program, which is categorized according to the approach used to solve the problem.
   a. Write and explain **one (1)** programming paradigm that you know.

_____

_____

b. Identify the suitable translator used by these programming languages.

```
0x52ac76:  movl   7306562(%ebx), %eax
0x52ac7c:  movl   %eax, -20(%ebp)
0x52ac7f:  movl   $0, (%edi,%eax)
0x52ac86:  testl  %esi, %esi
0x52ac88:  je     0x52ad21                ; -
    [UINavigationController _updateScrollViewFromViewCor
    toViewController:] + 425
0x52ac8e:  movl   7306542(%ebx), %eax
0x52ac94:  movl   (%edi,%eax), %eax
0x52ac97:  movl   %eax, -24(%ebp)
0x52ac9a:  movl   7212558(%ebx), %eax
0x52aca0:  movl   %eax, 4(%esp)
0x52aca4:  movl   %esi, (%esp)
0x52aca7:  calll  0x9bff06                ; symbol stu
```

Programming Language A

```
using namespace std;
nt main()
{
   int armstrong=0,num=0,result=0,check;
   cout<<"Enter a Number to find it armstrong or not"
   cin>>num;
   check=num;
   for(int i=1;num!=0;i++){
      armstrong=num%10;
      num=num/10;
      armstrong=armstrong*armstrong*armstrong;
      result=result+armstrong;
   }
```

Programming Language B

i. Translator Programming Language A:
ii. Translator Programming Language B:

c. Differentiate the translator mentioned in **8(c)(i)** and 8**(c)(ii).**

8. Programming language is a computer language engineered to create standard forms of commands that are used by humans to give instructions to computers. It can be categorized into two types; low level language and high level language.

   a. Explain one (1) difference between low level language and high level Language. [4 marks]

_____

_____

b. Most of the computer languages need a program named translator to assure the program could be executed successfully. State one (1) type of translator. [1 mark]

_____

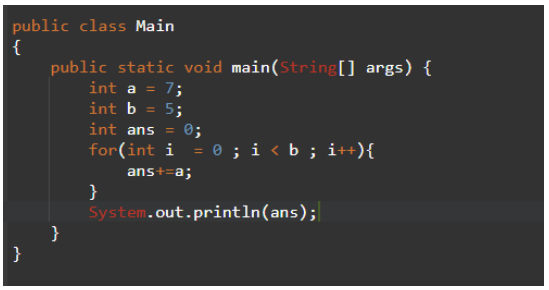
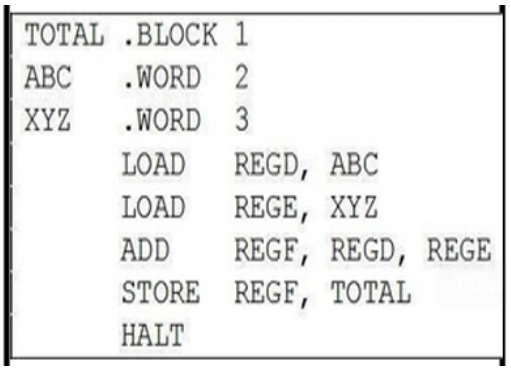
_____

c. Describe the function of the translator you have given in 4(b)(i). [2m]

_____

_____

9. Programming language is important because it defines the relationship, semantics and grammar which allows the programmers to effectively communicate with the machines that they program

Program A	**Program B**
	

```
public class Main
{
    public static void main(String[] args) {
        int a = 7;
        int b = 5;
        int ans = 0;
        for(int i  = 0 ; i < b ; i++){
            ans+=a;
        }
        System.out.println(ans);
    }
}
```

```
TOTAL .BLOCK 1
ABC   .WORD  2
XYZ   .WORD  3
      LOAD   REGD, ABC
      LOAD   REGE, XYZ
      ADD    REGF, REGD, REGE
      STORE  REGF, TOTAL
      HALT
```

a. Answer the following question based on above diagram

i. Identify the type of programming language (*1m*)

_____

ii. Give an example of translator that can be used to execute the Program B (*1m*)

_____

iii. State one difference between Program A and Program B. (*2m*)

_____